

Jin-Rou (Jennifer) Chang

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EDUCATION

University of Michigan

M.S. in Mechanical Engineering (GPA: 3.49/4.00)

Coursework: Design for Manufacturability; Design Optimization; Intro to Algorithmic Robotics; Vehicle Control Systems

Ann Arbor, MI

Aug 2025 – May 2027 (Est.)

National Taiwan University

B.S. in Mechanical Engineering (GPA: 3.70/4.00, Dean's List 2020)

Taipei, Taiwan

Sept. 2020 – June 2024

SKILLS

Mechanical Design & Analysis: SolidWorks, Inventor, AutoCAD, ABAQUS (FEA), CFD, GD&T, DFMEA

Robotics & Controls: MATLAB/Simulink, Python, C/C++, Arduino, OpenCV, motion planning (RRT), Git, Linux (basic)

Manufacturing & Quality: Machining (lathe, mill, grinding), welding, 3D printing, laser cutting

WORK EXPERIENCE

Garmin Ltd.

Intern, New Product Introduction Department, Product Engineering, SMT Team

New Taipei, Taiwan

July 2023 – June 2024

- Authored **Design for Manufacturability** documentation to support the transition from pilot runs to mass manufacturing, reducing rework costs and improving production readiness.
- Created 160+ laser-cut PCB fixture diagrams in **AutoCAD** for testing and validation purposes, enabling efficient fixture fabrication and verification across 8 products.
- Investigated electromechanical failures during PCB soldering processes and applied red dye penetration testing to identify welding defects, improving defect detection and manufacturing quality.

PROJECT EXPERIENCE

Vision-Guided Autonomous Block Stacking with RX200

Robotics Systems Lab, University of Michigan

Ann Arbor, MI

Jan. 2026 – Present

- Designed and integrated a vision-guided manipulation system combining an **Interbotix RX200 5-DOF arm** and **Intel RealSense L515 RGB-D camera** for autonomous block sorting and stacking.
- Implemented a geometry-based cube classification module in **OpenCV** using convex hull approximation, aspect ratio filtering, min-area bounding box fitting to reject triangular, cylindrical, and irregular blocks under RGB-D scenes.
- Developed and tested a closed-loop stacking routine using camera-derived world coordinates, iterating through lab debugging to improve reliability on real hardware.

Kinodynamic Motion Planning (RRT-based)

Introduction to Algorithmic Robotics, University of Michigan

Ann Arbor, MI

Sept. 2025 – Dec. 2025

- Implemented a kinodynamic RRT planner in **Python** for a planar hovercraft via forward simulation of system dynamics and motion primitives, enabling collision-free planning under acceleration and velocity constraints.
- Built a physics-based PyBullet simulation with obstacles and evaluated planner performance by varying motion primitives, visualizing the search tree and generating dynamically feasible trajectories that reached the goal.

Smoothie Maker Redesign with Manufacturability Analysis

Design for Manufacturing, University of Michigan

Ann Arbor, MI

Sept. 2025 – Dec. 2025

- Redesigned CAD modules in **SolidWorks** for a professional-use variant, increasing container capacity, adding handles for portability while preserving common mechanical interfaces to enable cross-variant interchangeability.
- Performed DFMEA-driven redesign of a stainless-steel sheet metal component using **SolidWorks DFMPro** and applied analytical cost models to evaluate manufacturability and process-dependent trade-offs for high-volume production.

Google Hardware Product Sprint - RFID-Enabled Laundry Sorting Basket

Selected Participant, Google (Industry Program)

New Taipei, Taiwan

June 2024 – Sept. 2024

- Created mechanical components in **Inventor** and **AutoCAD**, integrating **Arduino** control and fabricating prototypes via **3D printing** and **laser-cut** MDF and acrylic, resulting in a fully integrated and functional system prototype.
- Presented functional prototype and communicated design rationale and results in a 5-person team presentation, earning recognition from Google hardware engineers.